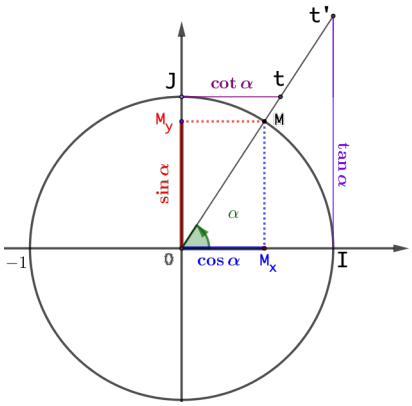
Formules trigonométriques

✯ Tableau de valeurs ✯

α	$0°$	$\frac{\pi}{6} = 30°$	$\frac{\pi}{4} = 45°$	$\frac{\pi}{3} = 60°$	$\frac{\pi}{2} = 90°$	$\pi = 180°$	$\frac{3\pi}{2} = 270°$
$\cos(\alpha)$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	-1	0
$\sin(\alpha)$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	0	-1
$\tan(\alpha)$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	$\|\|$	0	$\|\|$



✯ Transformations trigonométriques remarquables ✯

Angle $-\alpha$

(1) $\cos(-\alpha) = +\cos\alpha$
(2) $\sin(-\alpha) = -\sin\alpha$
(3) $\tan(-\alpha) = -\tan\alpha$

Angle $(\pi - \alpha)$

(4) $\cos(\pi - \alpha) = -\cos\alpha$
(5) $\sin(\pi - \alpha) = +\sin\alpha$
(6) $\tan(\pi - \alpha) = -\tan\alpha$

Angle $(\pi - \alpha)$

(7) $\cos(\pi + \alpha) = -\cos\alpha$
(8) $\sin(\pi + \alpha) = -\sin\alpha$
(9) $\tan(\pi + \alpha) = +\tan\alpha$

Angle $(\frac{\pi}{2} - \alpha)$

(10) $\cos\left(\frac{\pi}{2} - \alpha\right) = +\sin\alpha$
(11) $\sin\left(\frac{\pi}{2} - \alpha\right) = +\cos\alpha$
(12) $\tan\left(\frac{\pi}{2} - \alpha\right) = +\cot\alpha$

Angle $(\frac{\pi}{2} + \alpha)$

(13) $\cos\left(\frac{\pi}{2} + \alpha\right) = -\sin\alpha$
(14) $\sin\left(\frac{\pi}{2} + \alpha\right) = +\cos\alpha$
(15) $\tan\left(\frac{\pi}{2} + \alpha\right) = -\cot\alpha$

Periodicité

(16) $\cos(2k\pi + \alpha) = +\cos\alpha\,, \quad (k \in \mathbb{Z})$
(17) $\sin(2k\pi + \alpha) = +\sin\alpha\,, \quad (k \in \mathbb{Z})$
(18) $\tan(k\pi + \alpha) = +\tan\alpha\,, \quad (k \in \mathbb{Z})$

✯ Formules d'addition et de différence des arcs ✯

(19) $\sin(\alpha+\beta) = \sin\alpha\cos\beta + \sin\beta\cos\alpha$
(20) $\sin(\alpha-\beta) = \sin\alpha\cos\beta - \sin\beta\cos\alpha$
(21) $\tan(\alpha+\beta) = \dfrac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta}$
(22) $\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$
(23) $\cos(\alpha-\beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$
(24) $\tan(\alpha-\beta) = \dfrac{\tan\alpha - \tan\beta}{1 + \tan\alpha\tan\beta}$

Formules de produits en sommes

(25) $\sin\alpha\sin\beta = \frac{1}{2}\left[\cos(\alpha-\beta) - \cos(\alpha+\beta)\right]$
(26) $\cos\alpha\cos\beta = \frac{1}{2}\left[\cos(\alpha-\beta) + \cos(\alpha+\beta)\right]$
(27) $\tan\alpha\tan\beta = \dfrac{\tan\alpha + \tan\beta}{\cot\alpha + \cot\beta}$
(28) $\sin\alpha\cos\beta = \frac{1}{2}\left[\sin(\alpha+\beta) + \sin(\alpha-\beta)\right]$
(29) $\cot\alpha\cot\beta = \dfrac{\cot\alpha + \cot\beta}{\tan\alpha + \tan\beta}$
(30) $\tan\alpha\cot\beta = \dfrac{\tan\alpha + \cot\beta}{\cot\alpha + \tan\beta}$

Formules de sommes en produits

(31) $\sin\alpha + \sin\beta = 2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$
(32) $\sin\alpha - \sin\beta = 2\sin\left(\frac{\alpha-\beta}{2}\right)\cos\left(\frac{\alpha+\beta}{2}\right)$
(33) $\tan\alpha + \tan\beta = \dfrac{\sin(\alpha+\beta)}{\cos\alpha\cos\beta}$
(34) $\cos\alpha + \cos\beta = 2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$
(35) $\cos\alpha - \cos\beta = -2\sin\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right)$
(36) $\tan\alpha - \tan\beta = \dfrac{\sin(\alpha-\beta)}{\cos\alpha\cos\beta}$
(37) $\cos\alpha + \sin\alpha = \sqrt{2}\sin\left(\frac{\pi}{4} + \alpha\right)$
(38) $\cos\alpha - \sin\alpha = \sqrt{2}\sin\left(\frac{\pi}{4} - \alpha\right)$
(39) $\cot\alpha + \cot\beta = \dfrac{\sin(\alpha+\beta)}{\sin\alpha\sin\beta}$
(40) $\cot\alpha - \cot\beta = \dfrac{\sin(\alpha-\beta)}{\sin\alpha\sin\beta}$
(41) $\tan\alpha + \cot\beta = +\dfrac{\cos(\alpha-\beta)}{\cos\alpha\sin\beta}$
(42) $\tan\alpha - \cot\beta = -\dfrac{\cos(\alpha+\beta)}{\cos\alpha\sin\beta}$

Double-Angle Formulas

(43) $\sin(2\alpha) = 2\sin\alpha\cos\alpha$
(44) $\cos(2\alpha) = \cos^2\alpha - \sin^2\alpha = 2\cos^2\alpha - 1 = 1 - 2\sin^2\alpha$
(45) $\tan(2\alpha) = \dfrac{2\tan\alpha}{1 - \tan^2\alpha}$
(46) $\sin^2(\alpha) = \dfrac{1 - \cos(2\alpha)}{2}$
(47) $\cos^2(\alpha) = \dfrac{1 + \cos(2\alpha)}{2}$

✯ Équations trigonométriques ✯

Valeur des angles limites

(48) $\sin(x) = (-1)^k \iff x = \frac{\pi}{2} + k\pi$
(49) $\sin(x) = 0 \iff x = k\pi$
(50) $\cos(x) = 0 \iff x = \frac{\pi}{2} + k\pi$
(51) $\cos(x) = (-1)^k \iff x = k\pi$

Principe d'équivalence

(52) $\cos(x) = \cos(\alpha) \iff x = \begin{cases} \alpha + 2k\pi \\ -\alpha + 2k\pi \end{cases}$
(53) $\sin(x) = \sin(\alpha) \iff x = \begin{cases} \alpha + 2k\pi \\ \pi - \alpha + 2k\pi \end{cases}$
(54) $\tan(x) = \tan(\alpha) \iff x = \alpha + k\pi$
(55) $\cot(x) = \cot(\alpha) \iff x = \alpha + k\pi\,.$

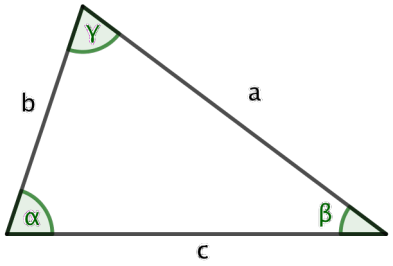
$\sin\left(n\frac{\pi}{2}\right)$

$$(56)\ \sin\left(n\frac{\pi}{2}\right)=\begin{cases}0, & n \text{ even}\\ 1, & n=1,5,9,\cdots\\ -1, & n=3,7,11,\cdots\end{cases}$$

$\cos\left(n\frac{\pi}{2}\right)$

$$(57)\ \cos\left(n\frac{\pi}{2}\right)=\begin{cases}0, & n \text{ odd}\\ 1, & n=0,4,8,\cdots\\ -1, & n=2,6,10,\cdots\end{cases}$$

✯ Lois des sinus, cosinus et tangente ✯



Aire du triangle

$$(58)\ \mathcal{A}=\frac{1}{2}ab\sin\gamma=\frac{1}{2}bc\sin\alpha=\frac{1}{2}ac\sin\beta$$

Loi de Sinus

$$(59)\ \frac{a}{\sin\alpha}=\frac{b}{\sin\beta}=\frac{c}{\sin\gamma}$$

Loi de Cosinus

$$(60)\ a^2=b^2+c^2-2bc\cos\alpha$$
$$(61)\ b^2=a^2+c^2-2ac\cos\beta$$
$$(62)\ c^2=b^2+a^2-2ba\cos\gamma$$

Loi de Tangente

$$(63)\ \frac{a-b}{a+b}=\frac{\tan\frac{1}{2}(\alpha-\beta)}{\tan\frac{1}{2}(\alpha+\beta)}$$
$$(64)\ \frac{b-c}{b+c}=\frac{\tan\frac{1}{2}(\beta-\gamma)}{\tan\frac{1}{2}(\beta+\gamma)}$$
$$(65)\ \frac{a-c}{a+c}=\frac{\tan\frac{1}{2}(\alpha-\gamma)}{\tan\frac{1}{2}(\alpha+\gamma)}$$

✯ Dérivée des fonctions trigonométriques ✯

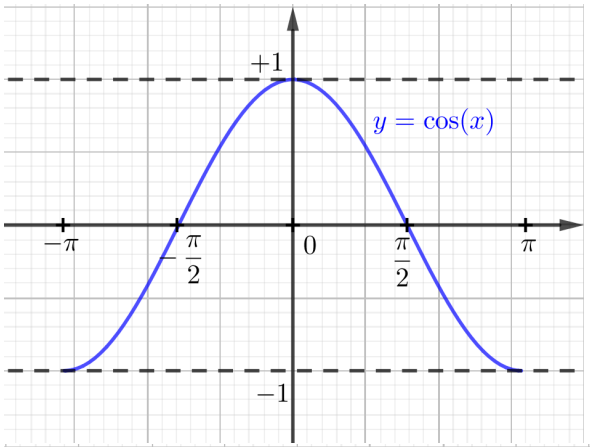
$$(66)\ \cos'\left(u(x)\right)=-u'(x)\sin\left(u(x)\right)\qquad (67)\ \sin'\left(u(x)\right)=u'(x)\cos\left(u(x)\right)\qquad (68)\ \tan'\left(u(x)\right)=\frac{u'(x)}{\cos^2\left(u(x)\right)}$$

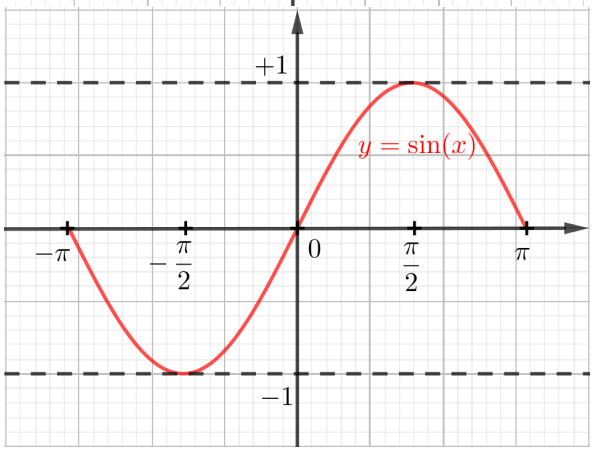
$$(69)\ \cot'\left(u(x)\right)=-\frac{u'(x)}{\sin^2\left(u(x)\right)}\qquad (70)\ \cos^{(n)}(x)=\cos\left(x+n\frac{\pi}{2}\right)\qquad (71)\ \sin^{(n)}(x)=\sin\left(x+n\frac{\pi}{2}\right)$$

✯ Représentation graphique ✯

x	$-\pi$		0		$+\pi$
$\cos'(x)$		$+$	0	$-$	
$\cos(x)$	-1	↗	1	↘	-1

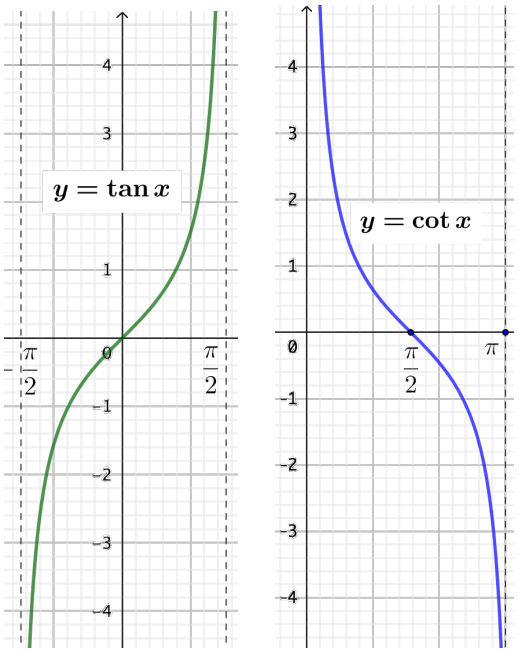
x	$-\pi$		$-\frac{\pi}{2}$		$\frac{\pi}{2}$		$+\pi$
$\sin'(x)$		$-$	0	$+$	0	$-$	
$\sin(x)$	0	↘	-1	↗	1	↘	0





x	$-\frac{\pi}{2}$		0		$\frac{\pi}{2}$
$\tan'(x)$	$+$		$+$		
$\tan(x)$	$-\infty$	↗	0	↗	$+\infty$

x	0		$\frac{\pi}{2}$		π
$\cot'(x)$	$-$		$-$		
$\cot(x)$	$+\infty$	↘	0	↘	$-\infty$



✯ Intégrale des fonctions trigonométriques ✯

Primitives

(72) $\int \cos x\,dx = \sin x$ (73) $\int \sin x\,dx = -\cos x$ (74) $\int \tan x\,dx = -\ln|\cos x|$ (75) $\int \cot x\,dx = -\ln|\sin x|$

Bioche rule $\int \omega(x)$ with $\omega(x) = f(x)\,dx$

(76) $\omega(-x) = \omega(x)$ take $t = \cos(x)$ (77) $\omega(\pi - x) = \omega(x)$ take $t = \sin(x)$
(78) $\omega(\pi + x) = \omega(x)$ take $t = \tan(x)$ (79) If 2 of 3 is true take $t = \cos(2x)$

(80) If all above are failed take $t = \tan(x/2)$

$$\cos x = \frac{1-t^2}{1+t^2} \quad \sin x = \frac{2t}{1+t^2}, \quad \tan x = \frac{2t}{1-t^2}, \quad \mathrm{d}x = 2\frac{dt}{1+t^2}$$

Integral of the form $\int \sin^m x \cos^n x\,dx$

(81) If $m = 2k+1$

take $t = \cos x$ then $\int \sin^m x \cos^n x\,dx = -\int t^n(1-t^2)^k\,dt$

(82) If $n = 2k+1$ with (m even)

take $t = \sin x$ then $\int \sin^m x \cos^n x\,dx = +\int t^m(1-t^2)^k\,dt$

(83) If m and v are even

$$\sin^2(x) = \frac{1-\cos(2x)}{2}, \quad \cos^2(x) = \frac{1+\cos(2x)}{2}, \quad \cos x \sin x = \frac{\sin(2x)}{2}$$

✯ Others ✯

Euler formula

(84) $\cos x = \dfrac{e^{ix}+e^{-ix}}{2}$

(85) $\sin x = \dfrac{e^{ix}-e^{-ix}}{2i}$

(86) $\tan x = i\dfrac{1-e^{2ix}}{1+e^{2ix}}$

(87) $\cot x = i\dfrac{1+e^{2ix}}{1-e^{2ix}}$

Finite expansion

(88) $\cos x = 1 - \dfrac{x^2}{2!} + \dfrac{x^4}{4!} + \cdots + (-1)^n \cdot \dfrac{x^{2n}}{(2n)!} + o(x^{2n+1})$

(89) $\sin x = x - \dfrac{x^3}{3!} + \dfrac{x^5}{5!} + \cdots + (-1)^n \cdot \dfrac{x^{2n+1}}{(2n+1)!} + o(x^{2n+2})$

(90) $\tan x = x + \dfrac{x^3}{3} + \dfrac{2}{15}x^5 + \dfrac{17}{315}x^7 + o(x^8)$

✯ Fonction trigonométrique réciproque ✯

$$\begin{array}{ccccc} \arcsin : & [-1;1] & \longmapsto & \left[-\frac{\pi}{2};\frac{\pi}{2}\right] \\ & y & \longmapsto & x=\arcsin y \end{array} \qquad \begin{array}{ccccc} \arccos : & [-1;1] & \longmapsto & [0;\pi] \\ & y & \longmapsto & x=\arccos y \end{array} \qquad \begin{array}{ccccc} \arctan : &]-\infty;+\infty[& \longmapsto & \left]-\frac{\pi}{2};\frac{\pi}{2}\right[\\ & y & \longmapsto & x=\arctan y \end{array}$$

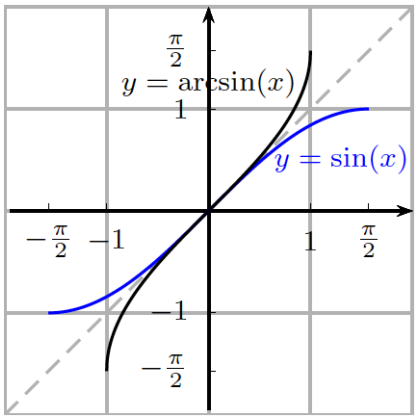
(91) $y=\cos x \Longleftrightarrow x=\arccos y$ (92) $y=\sin x \Longleftrightarrow x=\arcsin y$ (93) $y=\tan x \Longleftrightarrow x=\arctan y$

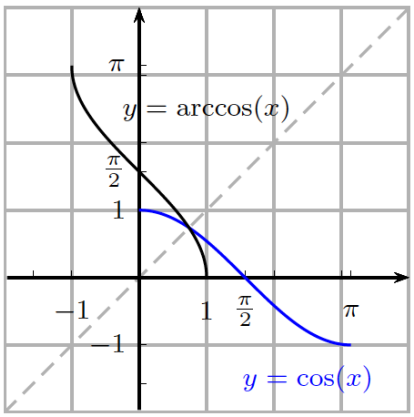
Tableau de variation et représentation graphique

x	$-\frac{\pi}{2}$		0		$\frac{\pi}{2}$
$\sin x$	-1	↗	0	↗	1
y	-1		0		1
$\arcsin y$	$-\frac{\pi}{2}$	↗	0	↗	$\frac{\pi}{2}$

x	0		$\frac{\pi}{2}$		π
$\cos x$	1	↘	0	↘	-1
y	-1		0		1
$\arccos y$	π	↘	$\frac{\pi}{2}$	↘	0

x	$-\frac{\pi}{2}$		0		$\frac{\pi}{2}$
$\tan x$	$-\infty$	↗	0	↗	$+\infty$
y	$-\infty$		0		$+\infty$
$\arctan y$	$-\frac{\pi}{2}$	↗	0	↗	$+\frac{\pi}{2}$





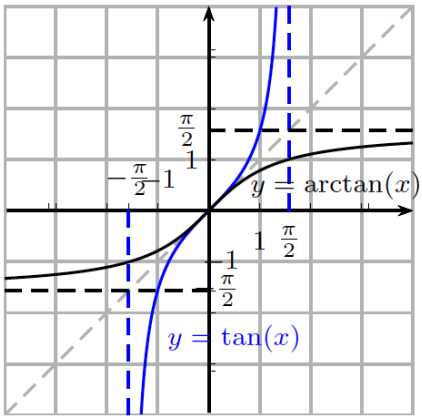


Tableau de valeurs pour les fonctions trigonométiques réciproques

x	-1	$-\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{2}}{2}$	$-\frac{1}{2}$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\arcsin x$	$-\frac{\pi}{2}=-90^0$	$-\frac{\pi}{3}=-60^0$	$-\frac{\pi}{4}=-45^0$	$-\frac{\pi}{6}=-30^0$	0^0	$\frac{\pi}{6}=30^0$	$\frac{\pi}{4}=45^0$	$\frac{\pi}{3}=60^0$	$\frac{\pi}{2}=90^0$
$\arccos x$	$\pi=180^0$	$\frac{5\pi}{6}=150^0$	$\frac{3\pi}{4}=135^0$	$\frac{2\pi}{3}=120^0$	$\frac{\pi}{2}=90^0$	$\frac{\pi}{3}=60^0$	$\frac{\pi}{4}=45^0$	$\frac{\pi}{6}=30^0$	0^0
x	$-\infty$	-1	$-\frac{\sqrt{3}}{3}$	$-\sqrt{3}$	0	$\sqrt{3}$	$\frac{\sqrt{3}}{3}$	1	$+\infty$
$\arctan x$	$-\frac{\pi}{2}=-90^0$	$-\frac{\pi}{4}=-45^0$	$-\frac{\pi}{6}=-30^0$	$-\frac{\pi}{3}=-60^0$	0^0	$\frac{\pi}{3}=60^0$	$\frac{\pi}{6}=30^0$	$\frac{\pi}{4}=45^0$	$\frac{\pi}{2}=90^0$
$\operatorname{arccot} x$	$-\pi=-180^0$	$\frac{3\pi}{4}=135^0$	$\frac{2\pi}{3}=120^0$	$\frac{5\pi}{6}=150^0$	$\frac{\pi}{2}=90^0$	$\frac{\pi}{6}=30^0$	$\frac{\pi}{3}=60^0$	$\frac{\pi}{4}=45^0$	0^0

Propriétés

(94) $\sin(\arcsin x)=x$ (98) $\arccos(-x)=\pi-\arccos x$ (102) $\sin(\arccos x)=\sqrt{1-x^2}$

(95) $\cos(\arccos x)=x$ (99) $\arcsin(-x)=-\arcsin x$ (103) $\cos(\arcsin x)=\sqrt{1-x^2}$

(96) $\tan(\arctan x)=x$ (100) $\arctan(-x)=-\arctan x$ (104) $\tan(\operatorname{arccot} x)=\frac{1}{x}$

(97) $\cot(\operatorname{arccot} x)=x$ (101) $\operatorname{arccot}(-x)=\pi-\operatorname{arccot} x$ (105) $\cot(\arctan x)=\frac{1}{x}$

Derivées

(106) $\arcsin'(x)=\dfrac{1}{\sqrt{1-x^2}}$ (107) $\arccos'(x)=-\dfrac{1}{\sqrt{1-x^2}}$ (108) $\arctan'(x)=\dfrac{1}{1+x^2}$ (109) $\operatorname{arccot}'(x)=-\dfrac{1}{1+x^2}$

(110) $\arcsin'\left(u(x)\right)=\dfrac{u'(x)}{\sqrt{1-u^2(x)}}$ (111) $\arccos'\left(u(x)\right)=-\dfrac{u'(x)}{\sqrt{1-u^2(x)}}$ (112) $\arctan'\left(u(x)\right)=\dfrac{u'(x)}{1+u^2(x)}$